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DSA0602 Data Handling and Visualization - Slot A

Unit II - Class Activity 2

Date: 24.07.2026

R code for all the graphs using employee salary dataset:

```
packages <- c("ggplot2","dplyr","ggridges","treemapify","hexbin","viridis")
```

```
for(pkg in packages){
```

```
  if(!require(pkg, character.only = TRUE)){
```

```
    install.packages(pkg)
```

```
    library(pkg, character.only = TRUE)
```

```
  }
```

```
}
```

```
library(ggplot2)
```

```
library(dplyr)
```

```
library(ggridges)
```

```
library(treemapify)
```

```
library(hexbin)
```

```
library(viridis)
```

```
df <- read.csv(file.choose(), stringsAsFactors = FALSE)
```

```
str(df)
```

```
df$Department <- as.factor(df$Department)
```

```
df$Gender <- as.factor(df$Gender)
df$Education_Level <- as.factor(df$Education_Level)
df$City <- as.factor(df$City)

theme_set(
  theme_minimal(base_size = 14) +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5),
    axis.title = element_text(face = "bold")
  )
)

ggplot(df, aes(x = Department, fill = Department)) +
  geom_bar() +
  scale_fill_viridis_d() +
  labs(
    title = "Employees by Department",
    x = "Department",
    y = "Count"
  ) +
  theme(legend.position = "none")

ggplot(df, aes(x = Monthly_Salary)) +
  geom_histogram(
    bins = 20,
    fill = "steelblue",
    color = "black"
```

```
) +  
labs(  
  title = "Distribution of Monthly Salary",  
  x = "Monthly Salary",  
  y = "Frequency"  
) +  
theme_minimal()
```

```
ggplot(df, aes(x = Monthly_Salary, fill = Gender)) +  
  geom_density(alpha = 0.5) +  
  scale_fill_viridis_d() +  
  labs(  
    title = "Density Plot of Monthly Salary by Gender",  
    x = "Monthly Salary",  
    y = "Density",  
    fill = "Gender"  
) +  
theme_minimal()
```

```
ggplot(df, aes(x = Department, y = Monthly_Salary, fill = Department)) +  
  geom_boxplot() +  
  scale_fill_viridis_d() +  
  labs(  
    title = "Monthly Salary by Department",  
    x = "Department",  
    y = "Monthly Salary",
```

```
    fill = "Department"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    legend.position = "none"
  )
```

```
ggplot(df, aes(x = Department, y = Monthly_Salary, fill = Department)) +
  geom_violin(trim = FALSE, alpha = 0.8) +
  geom_boxplot(width = 0.1, fill = "white", outlier.color = "red") +
  scale_fill_viridis_d() +
  labs(
    title = "Distribution of Monthly Salary by Department",
    x = "Department",
    y = "Monthly Salary"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    axis.title = element_text(face = "bold"),
    legend.position = "none"
  )
```

```
# Load package if not already loaded
```

```
library(ggribes)
```

```

ggplot(df,
  aes(x = Monthly_Salary,
      y = Department,
      fill = Department)) +
  geom_density_ridges(
    alpha = 0.8,
    scale = 1.2,
    color = "white"
  ) +
  scale_fill_viridis_d() +
  labs(
    title = "Ridgeline Plot of Monthly Salary by Department",
    x = "Monthly Salary",
    y = "Department"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    axis.title = element_text(face = "bold"),
    legend.position = "none"
  )

```

```

# Create summary data

```

```

pie_data <- df %>%

```

```

  count(Gender)

```

```
# Draw Pie Chart
```

```
ggplot(pie_data, aes(x = "", y = n, fill = Gender)) +  
  geom_col(width = 1, color = "white") +  
  coord_polar(theta = "y") +  
  scale_fill_viridis_d() +  
  labs(  
    title = "Gender Distribution of Employees",  
    fill = "Gender"  
  ) +  
  theme_void() +  
  theme(  
    plot.title = element_text(hjust = 0.5, face = "bold", size = 16),  
    legend.position = "right"  
  )
```

```
ggplot(df, aes(x = Department, fill = Gender)) +  
  geom_bar(color = "black") +  
  scale_fill_viridis_d() +  
  labs(  
    title = "Department-wise Employee Distribution by Gender",  
    x = "Department",  
    y = "Number of Employees",  
    fill = "Gender"  
  ) +  
  theme_minimal() +
```

```
theme(  
  plot.title = element_text(hjust = 0.5, face = "bold", size = 16),  
  axis.title = element_text(face = "bold"),  
  axis.text.x = element_text(angle = 45, hjust = 1),  
  legend.position = "right"  
)
```

```
install.packages("treemapify") # Only if not installed
```

```
library(treemapify)
```

```
# Count employees in each department
```

```
tree_data <- df %>%
```

```
  count(Department)
```

```
# Draw Tree Map
```

```
ggplot(tree_data,
```

```
  aes(  
    area = n,
```

```
    fill = Department,
```

```
    label = paste(Department, "\n", n)
```

```
  )) +
```

```
geom_treemap() +
```

```
geom_treemap_text(  
  colour = "white",
```

```
  place = "centre",
```

```
  grow = TRUE
```

```
) +
```

```
scale_fill_viridis_d() +  
labs(  
  title = "Employee Distribution by Department"  
) +  
theme_minimal() +  
theme(  
  plot.title = element_text(  
    hjust = 0.5,  
    face = "bold",  
    size = 16  
  ),  
  legend.position = "none"  
)
```

```
ggplot(df,  
  aes(x = Experience_Years,  
    y = Monthly_Salary,  
    color = Department)) +  
geom_point(size = 3, alpha = 0.8) +  
scale_color_viridis_d() +  
labs(  
  title = "Relationship Between Experience and Monthly Salary",  
  x = "Experience (Years)",  
  y = "Monthly Salary",  
  color = "Department"  
) +
```



```
theme_minimal() +  
theme(  
  plot.title = element_text(hjust = 0.5, face = "bold", size = 16),  
  axis.title = element_text(face = "bold"),  
  legend.position = "right"  
)
```

```
install.packages("hexbin") # Only if not installed
```

```
library(hexbin)
```

```
ggplot(df, aes(x = Age, y = Monthly_Salary)) +  
  geom_hex(bins = 20) +  
  scale_fill_viridis_c(option = "C") +  
  labs(  
    title = "Hexbin Plot of Age vs Monthly Salary",  
    x = "Age",  
    y = "Monthly Salary",  
    fill = "Count"  
  ) +
```

```
theme_minimal() +  
theme(  
  plot.title = element_text(hjust = 0.5, face = "bold", size = 16),  
  axis.title = element_text(face = "bold"),  
  legend.position = "right"  
)
```

```
ggplot(df, aes(x = Age, y = Monthly_Salary)) +
```

```

stat_density_2d(
  aes(fill = after_stat(level)),
  geom = "polygon",
  alpha = 0.7
) +
scale_fill_viridis_c() +
labs(
  title = "Filled 2D Density Plot of Age vs Monthly Salary",
  x = "Age",
  y = "Monthly Salary",
  fill = "Density"
) +
theme_minimal() +
theme(
  plot.title = element_text(hjust = 0.5, face = "bold", size = 16),
  axis.title = element_text(face = "bold")
)

```

```

ggplot(df,
  aes(x = Age,
    y = Monthly_Salary,
    size = Experience_Years,
    color = Department)) +
geom_point(alpha = 0.7) +
scale_color_viridis_d() +
scale_size(range = c(3, 10)) +

```

```

labs(
  title = "Bubble Chart: Age vs Monthly Salary",
  x = "Age",
  y = "Monthly Salary",
  size = "Experience (Years)",
  color = "Department"
) +
theme_minimal() +
theme(
  plot.title = element_text(hjust = 0.5, face = "bold", size = 16),
  axis.title = element_text(face = "bold"),
  axis.text = element_text(size = 12),
  legend.position = "right"
)

```

Calculate average salary for each experience level

```

line_data <- df %>%
  group_by(Experience_Years) %>%
  summarise(
    Average_Salary = mean(Monthly_Salary),
    .groups = "drop"
  )

```

Draw Line Graph

```

ggplot(line_data,
  aes(x = Experience_Years,

```

```

    y = Average_Salary)) +
geom_line(color = "steelblue", linewidth = 1.2) +
geom_point(color = "red", size = 3) +
labs(
  title = "Average Monthly Salary by Experience",
  x = "Experience (Years)",
  y = "Average Monthly Salary"
) +
theme_minimal() +
theme(
  plot.title = element_text(hjust = 0.5, face = "bold", size = 16),
  axis.title = element_text(face = "bold"),
  axis.text = element_text(size = 12)
)

```

```

ggplot(df,
  aes(x = Department,
    y = Monthly_Salary,
    color = Department)) +
geom_dotplot(
  binaxis = "y",
  stackdir = "center",
  dotsize = 0.7
) +
scale_color_viridis_d() +
labs(

```

```
title = "Dot Plot of Monthly Salary by Department",  
x = "Department",  
y = "Monthly Salary"  
) +  
theme_minimal() +  
theme(  
  plot.title = element_text(hjust = 0.5, face = "bold", size = 16),  
  axis.title = element_text(face = "bold"),  
  axis.text = element_text(size = 12),  
  legend.position = "none"  
)
```

```
# Create summary data
```

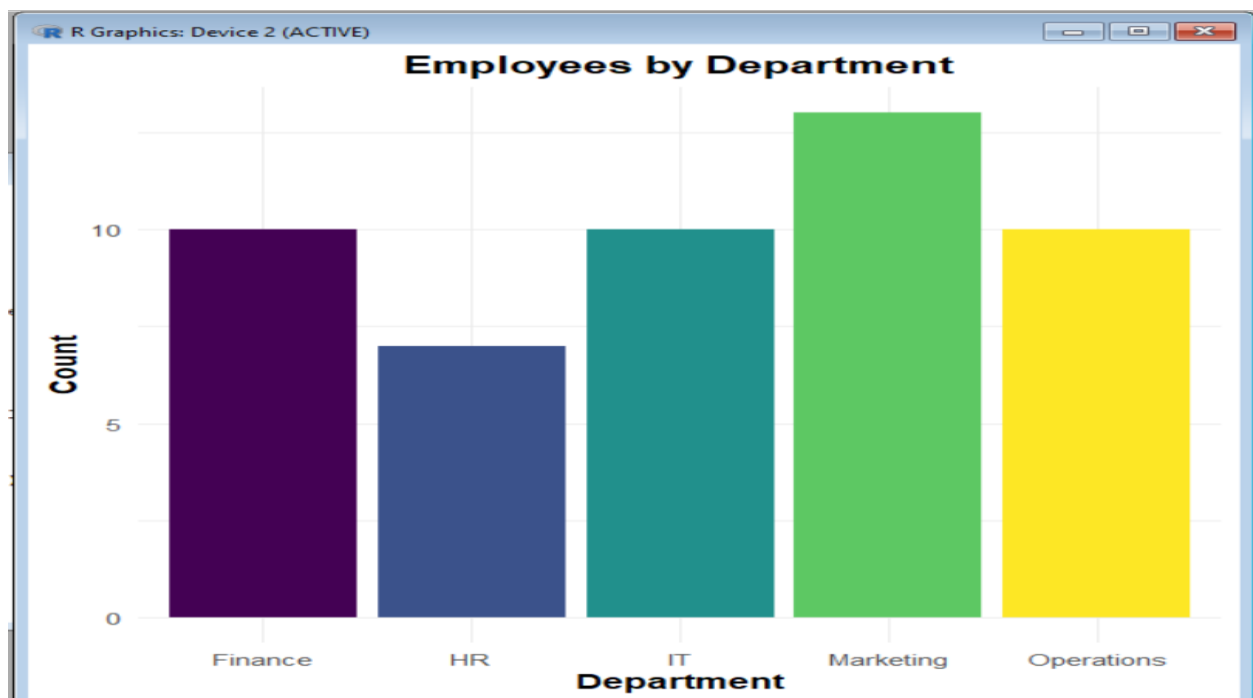
```
heat_data <- df %>%  
  count(Department, Education_Level)
```

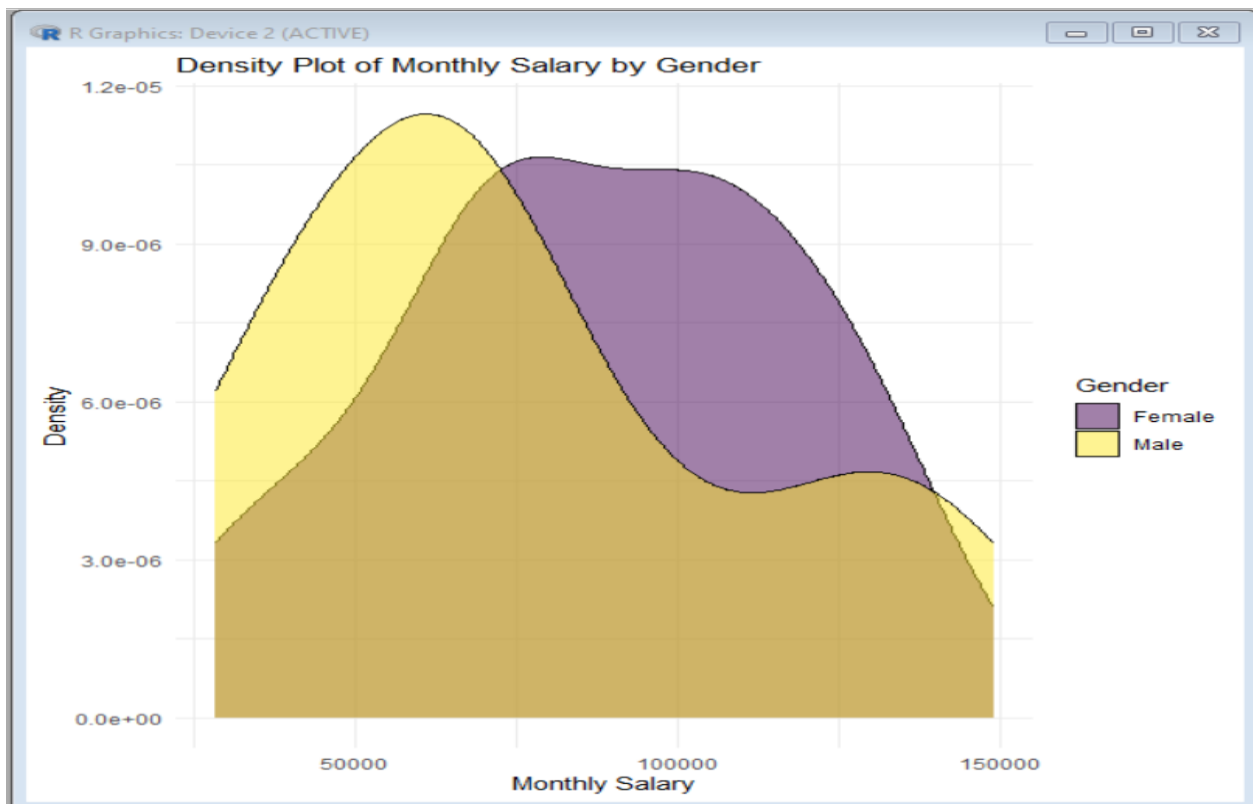
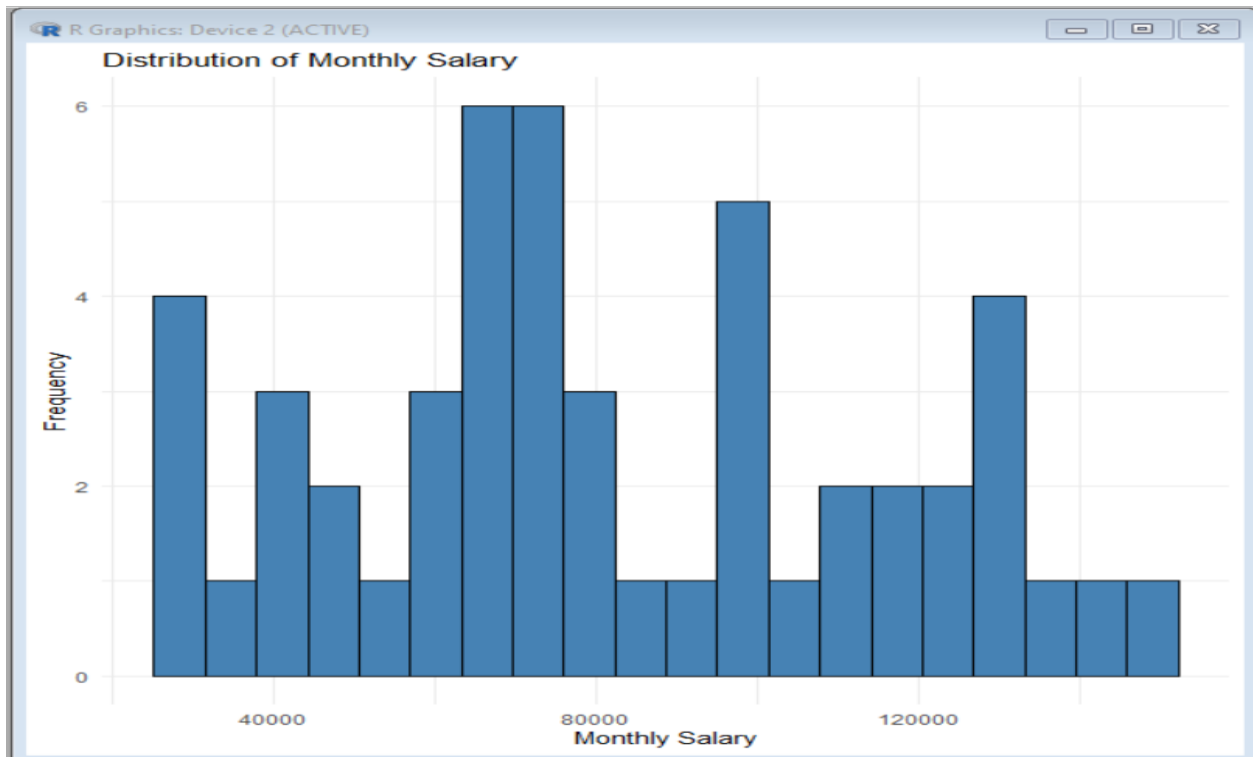
```
# Draw Heat Map
```

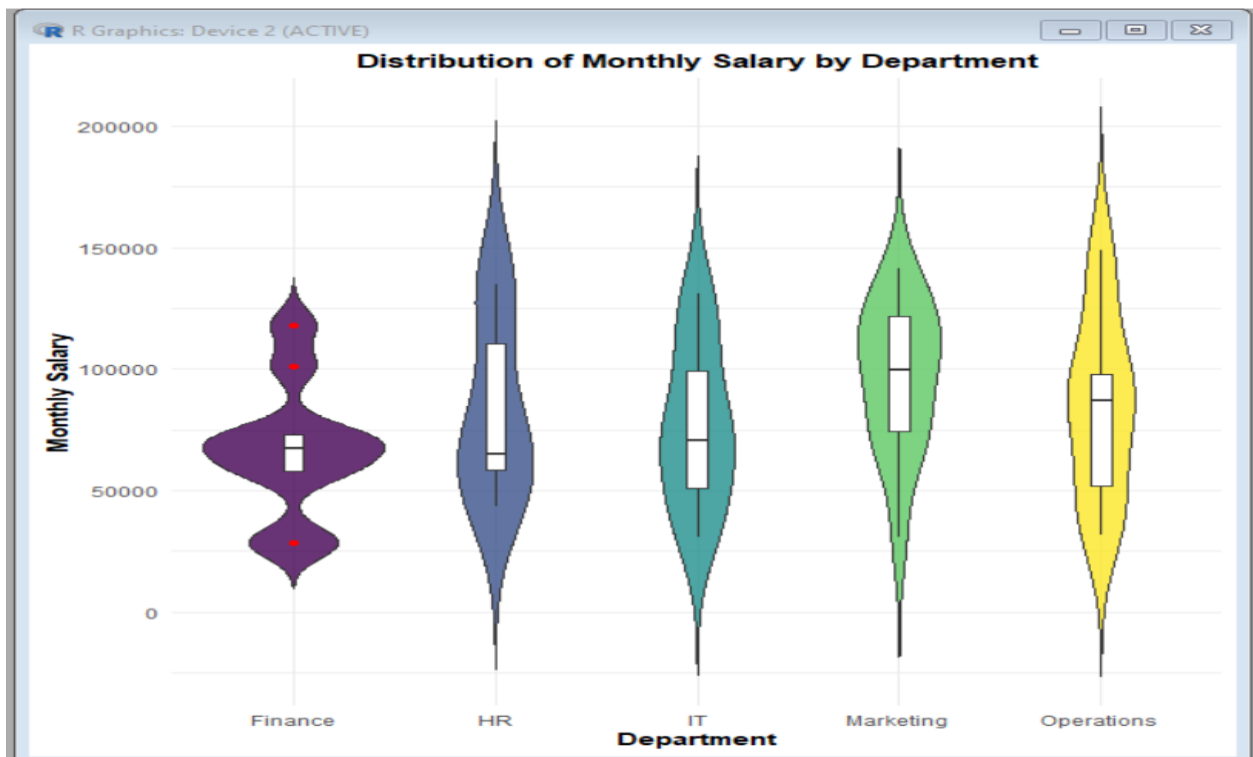
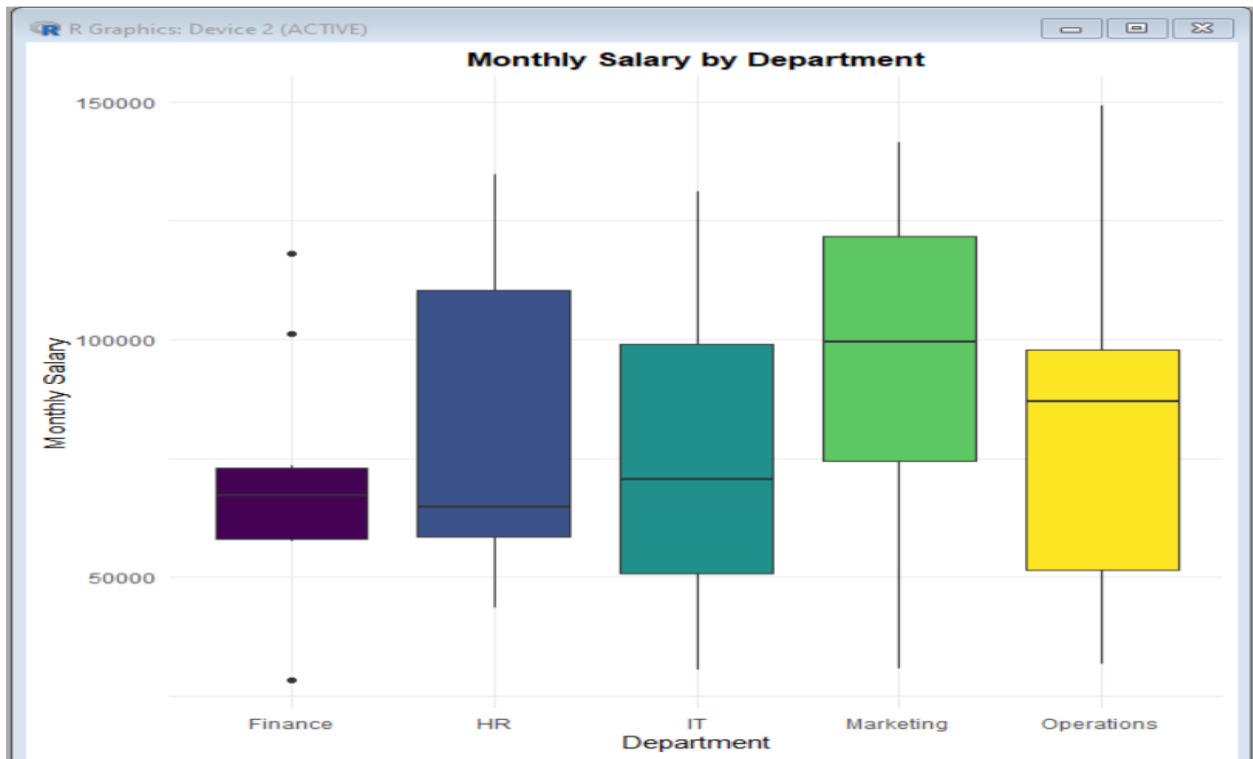
```
ggplot(heat_data,  
  aes(x = Department,  
    y = Education_Level,  
    fill = n)) +  
  geom_tile(color = "white") +  
  geom_text(aes(label = n), color = "black", size = 4) +  
  scale_fill_viridis_c() +  
  labs(  
    title = "Heat Map of Department vs Education Level",
```

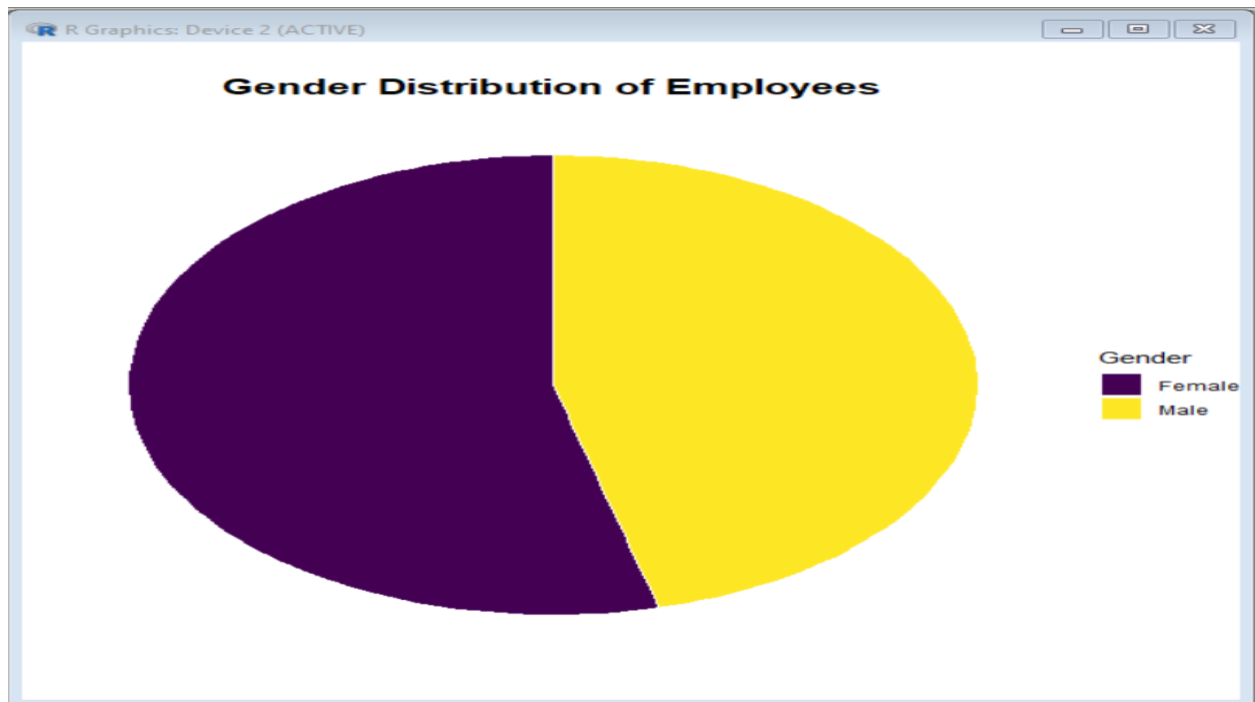
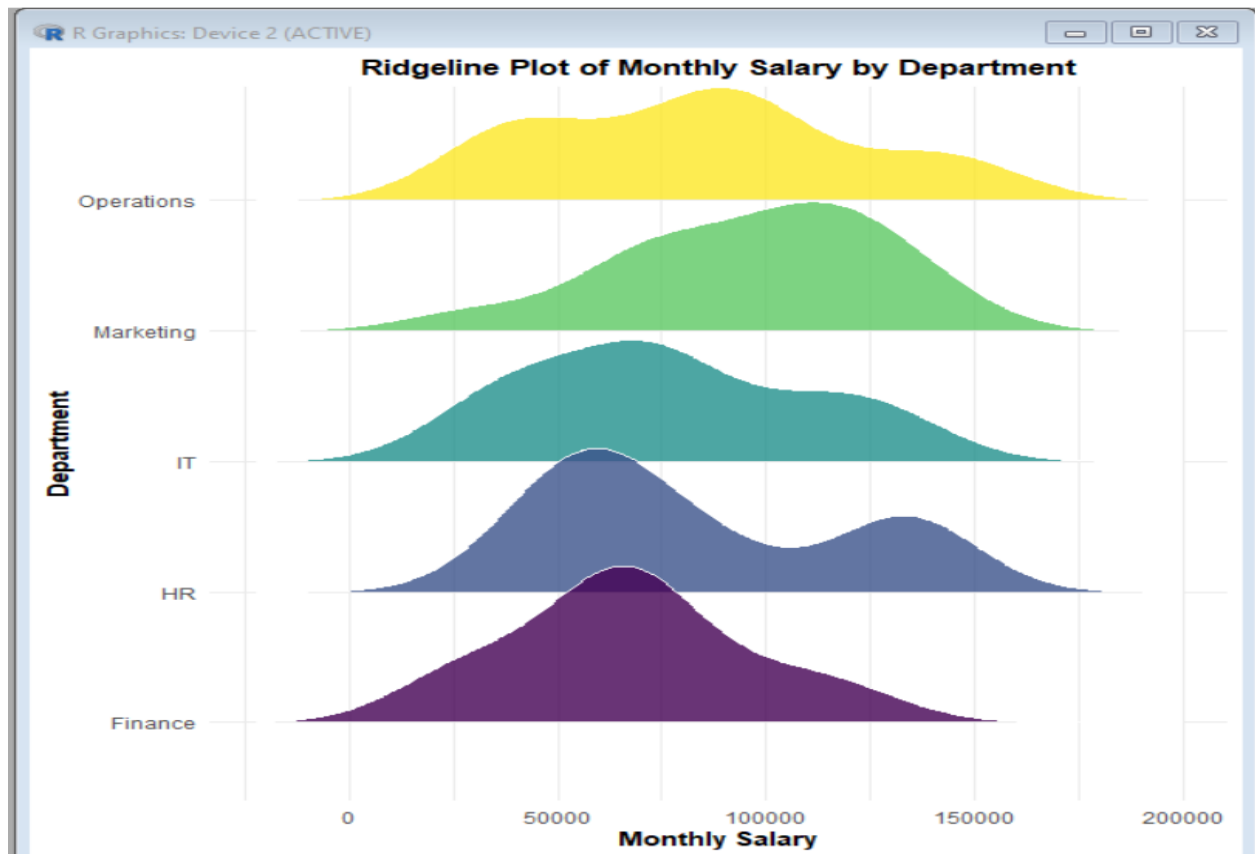
```
x = "Department",
y = "Education Level",
fill = "Employee Count"
) +
theme_minimal() +
theme(
  plot.title = element_text(hjust = 0.5, face = "bold", size = 16),
  axis.title = element_text(face = "bold"),
  axis.text.x = element_text(angle = 45, hjust = 1),
  axis.text = element_text(size = 11)
)
```

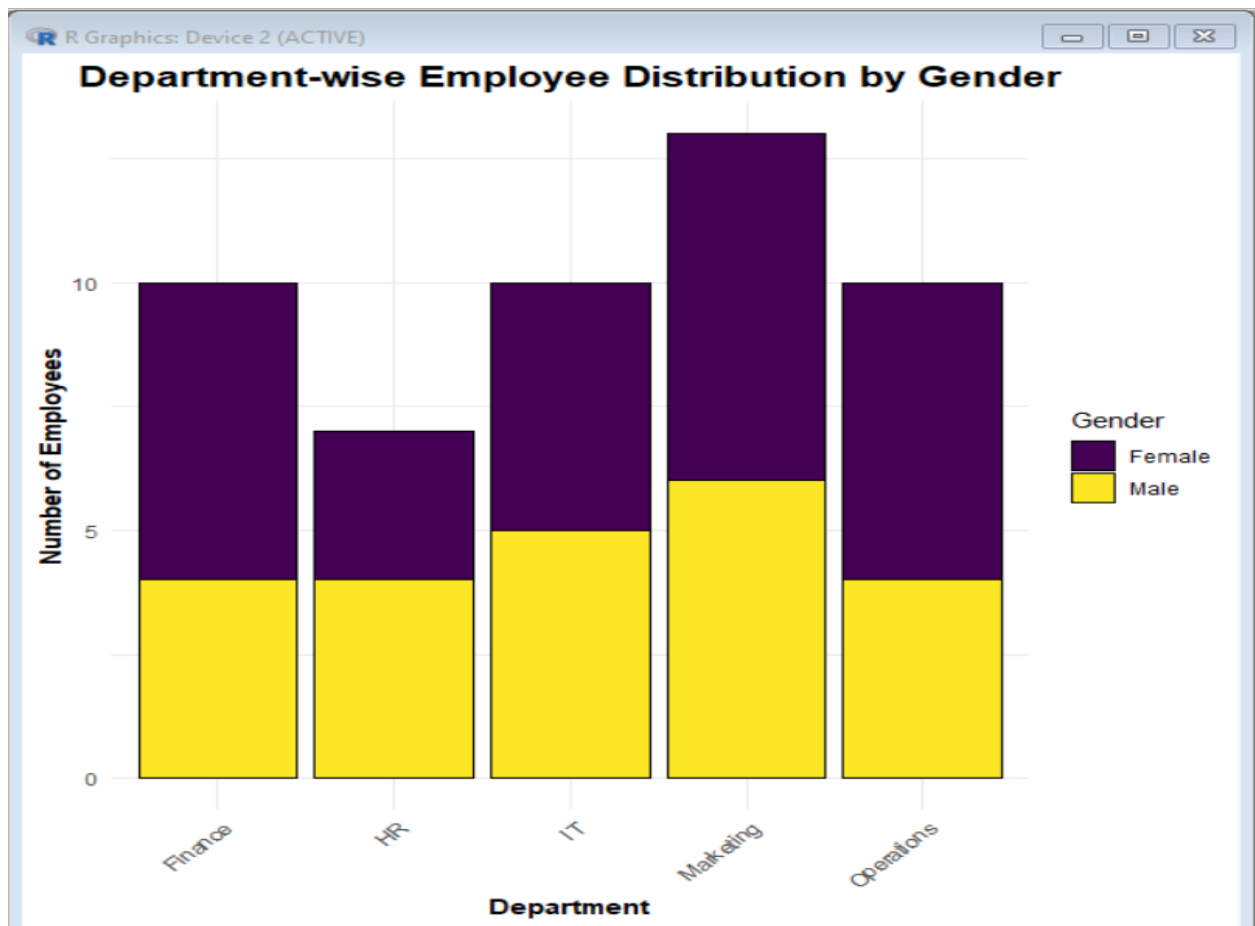
Outputs:

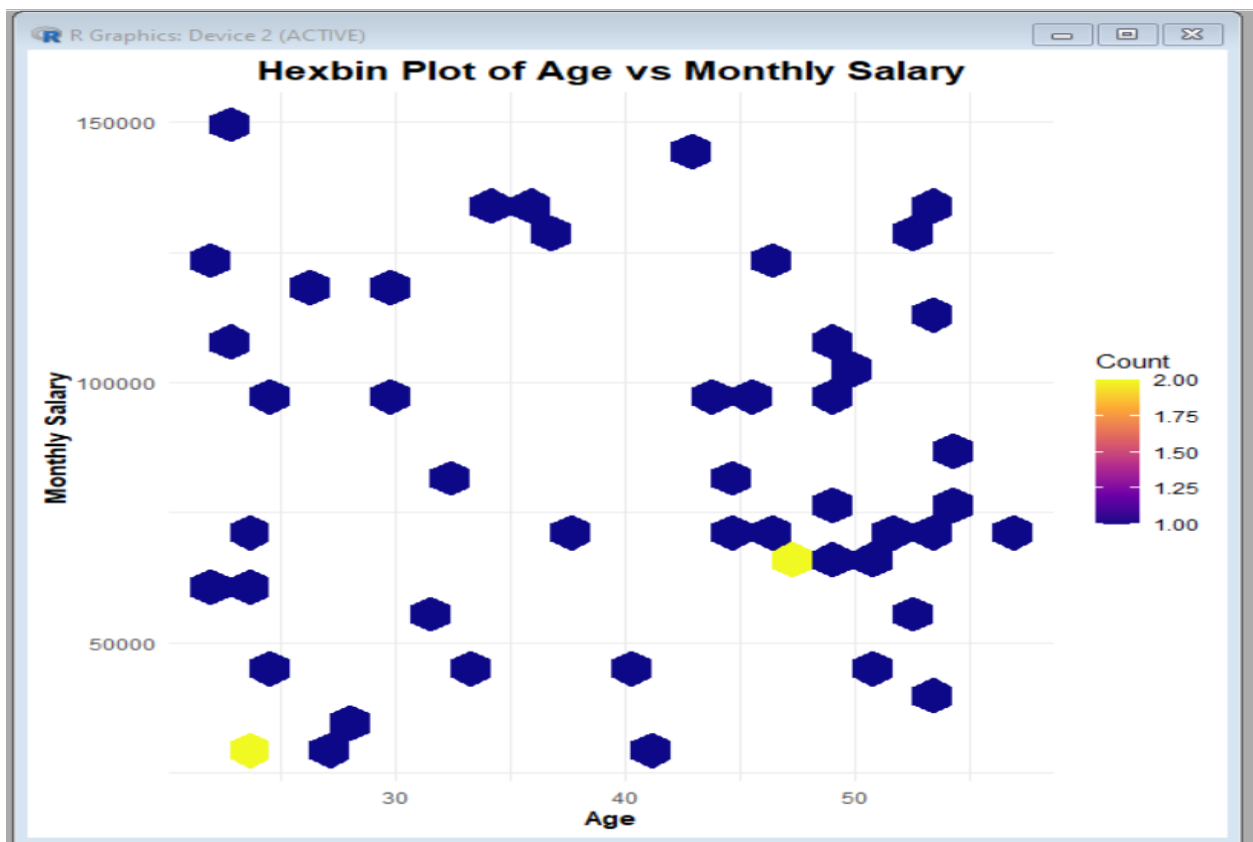
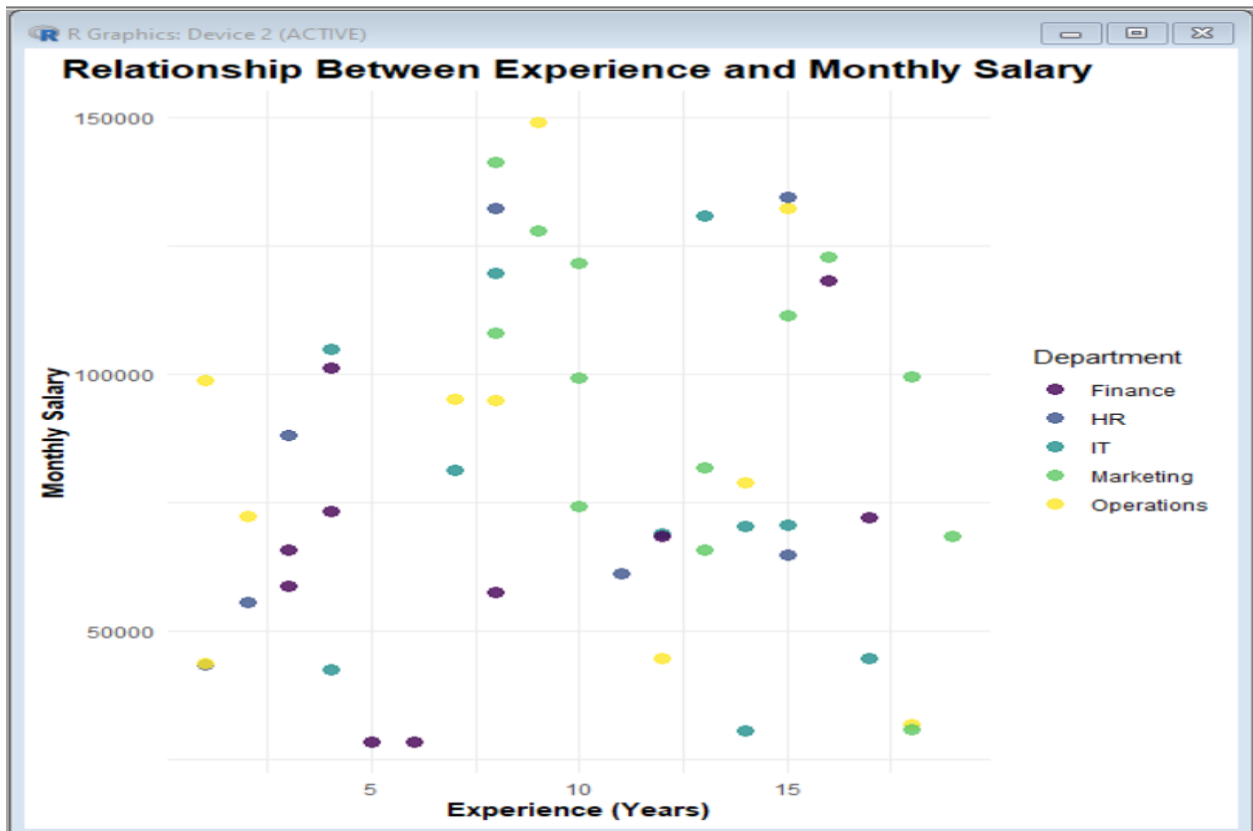


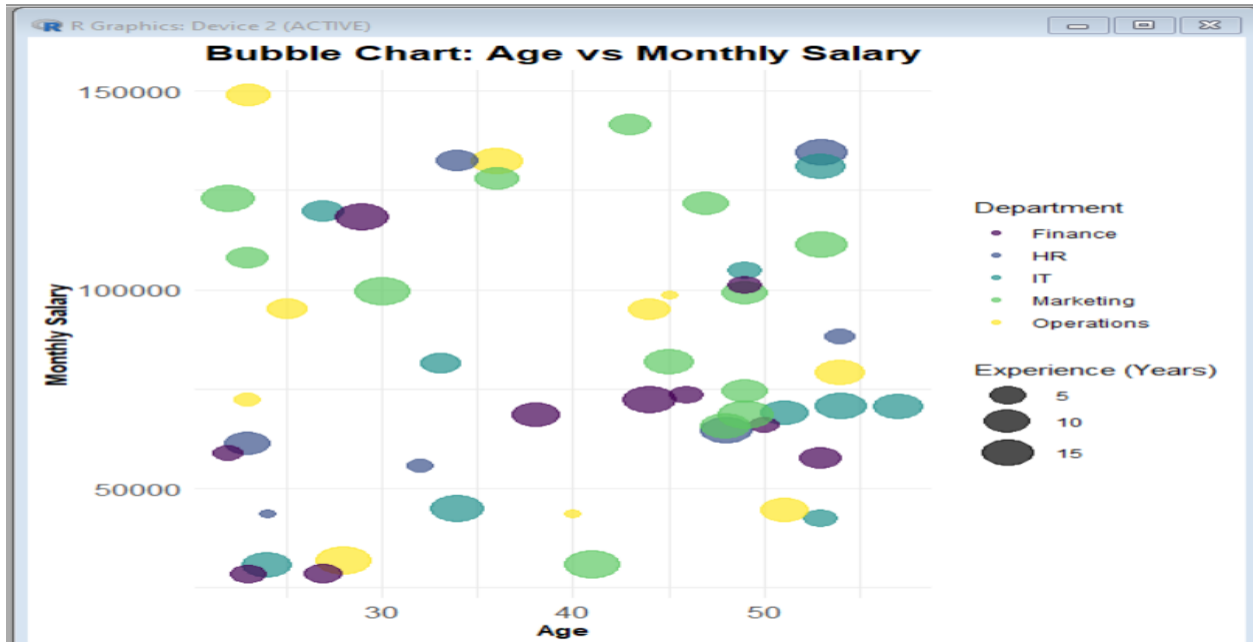
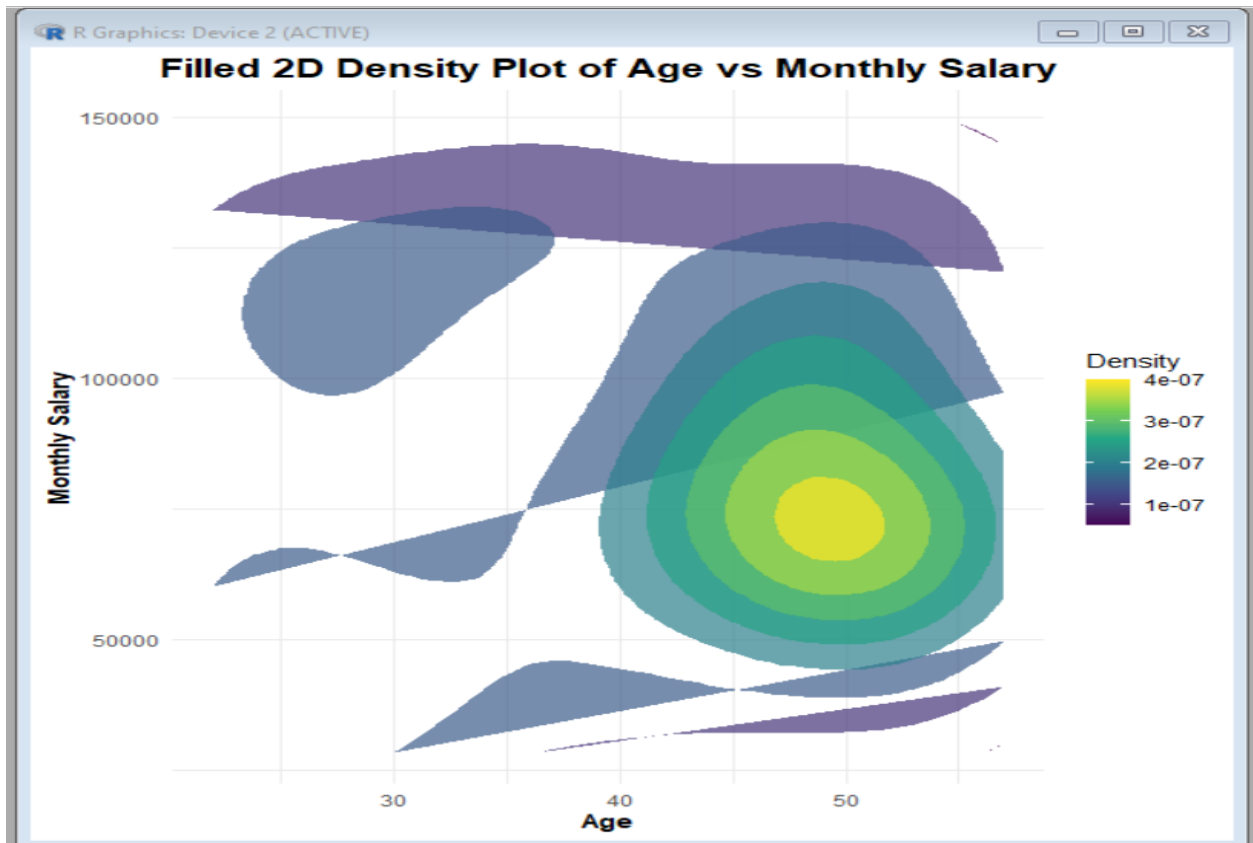












Heat Map of Department vs Education Level

